

Group#4

Deep Learning

(COSC-4121)

GoogleNet (InceptionNetwork)

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A: Methodology

1. Introduction to GoogLeNet

GoogLeNet, also known as Inception v1, is a deep convolutional neural network (CNN) that introduced the revolutionary **Inception Module**. Its primary strength is its ability to perform "multi-scale feature extraction." Instead of using a single filter size, it runs 1*1,3*3 and 5*5 convolutions in parallel. This allows the model to recognize both small details and large patterns in an image simultaneously. A key feature is the use of 1*1 convolutions for dimensionality reduction, which keeps the model computationally efficient and "lightweight" compared to other deep networks.

2. Description of Datasets

CIFAR-10: This dataset is a collection of 60,000 tiny 32*32 color images categorized into 10 classes, such as airplanes, birds, and ships. It is widely used to test a model’s ability to recognize general objects in low-resolution environments.

FashionMNIST: This dataset consists of 60,000 grayscale images of 10 clothing categories, such as sneakers, coats, and trousers. It serves as a more complex alternative to the standard MNIST digit dataset, requiring the model to capture intricate textures and shapes in apparel.

1. General Hyperparameters

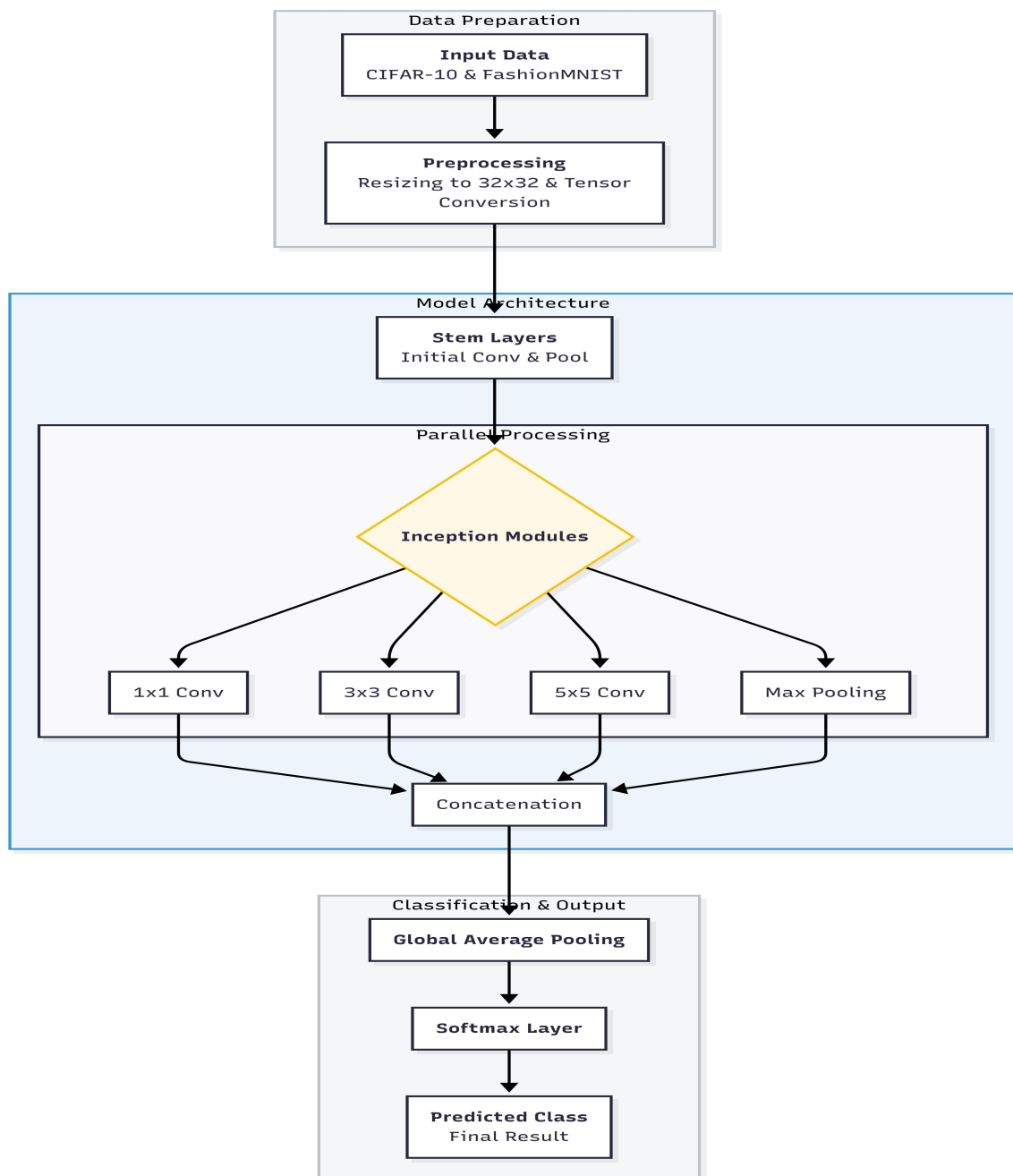
Hyperparameter	Description
Learning Rate	Determines the speed at which the model updates its weights during training.
Batch Size	The number of images processed before the model updates its internal parameters.
Epochs	The number of times the model sees the entire training dataset.
Optimizer	The algorithm (e.g., Adam) that adjusts weights to reduce the error/loss.

2. Experimental Hyperparameters

Table 1: Configuration of hyperparameters across the three experimental trials.

Experiment	Learning Rate	Batch Size	Epochs	Optimizer
Experiment 01	0.001	32	5	Adam
Experiment 02	0.0005	64	7	Adam
Experiment 03	0.0001	128	10	Adam

3. Methodology Diagram



Explanation

The proposed methodology follows a structured deep learning pipeline, beginning with a Data Preparation phase where CIFAR-10 and FashionMNIST images are resized to 32*32 and converted into tensors. The architecture then passes the data through the Stem Layers for initial spatial reduction and feature extraction. The heart of the process is the Inception Stack, which employs a "split-transform-merge" strategy; here, four parallel branches perform 1*1 , 3*3 and 5*5 convolutions

simultaneously to capture multi-scale features. After these features are concatenated, the model utilizes Global Average Pooling to flatten the data while minimizing the parameter count to prevent overfitting. Finally, the Softmax Layer calculates class probabilities to determine the most likely Output category.

B: Experimentation

1. Configuration and Tools

System Specifications: The experiments were executed on a laptop with an AMD Ryzen 5 7430U processor (2.30 GHz) and 8.00 GB RAM. The system runs on a 64-bit Windows OS.

Software Environment: The code was developed and executed using Python.

Dataset Source: The CIFAR-10 and FashionMNIST datasets were sourced from the official Keras/Torchvision repositories.

2. Results Table (Comparison)

Table 2: Comparative Results for CIFAR-10 and FashionMNIST.

Metric	CIFAR-10	FashionMNIST
Accuracy	14.2%	81.08%
Precision	0.06	0.82
Recall	0.14	0.81
F1-Score	0.08	0.81
ROC-AUC	0.50	0.50

3. Discussion of Graphs

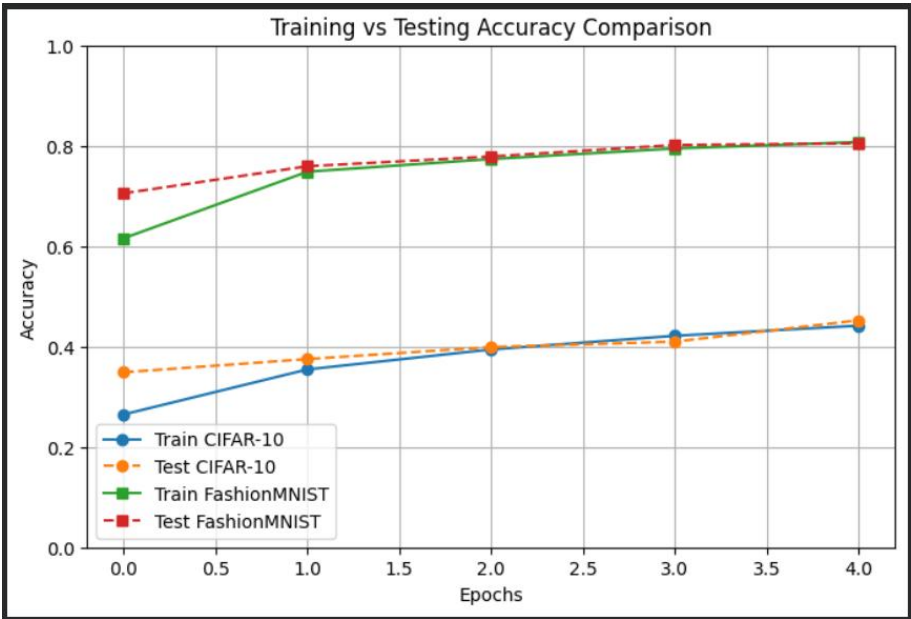


Figure 1: Training vs Testing Accuracy Comparison for both datasets

The learning trajectories depicted in Figure 1 reveal a high degree of proficiency in

the model's ability to classify grayscale imagery compared to RGB color data. For the FashionMNIST dataset, the curves indicate a robust learning phase, starting at a 60% baseline and converging near 80% with a very narrow gap between training and testing accuracy. This signifies excellent model generalization and a low risk of overfitting.

In contrast, the CIFAR-10 curves demonstrate a much lower performance ceiling, concluding at approximately 45%. The wider separation between the training and testing lines for CIFAR-10 highlights the challenges the simplified GoogLeNet architecture faces when processing the high-variance features of colored objects. These results logically support the conclusion that while the Inception modules are efficient, the complexity of the input data remains a primary driver of final accuracy.

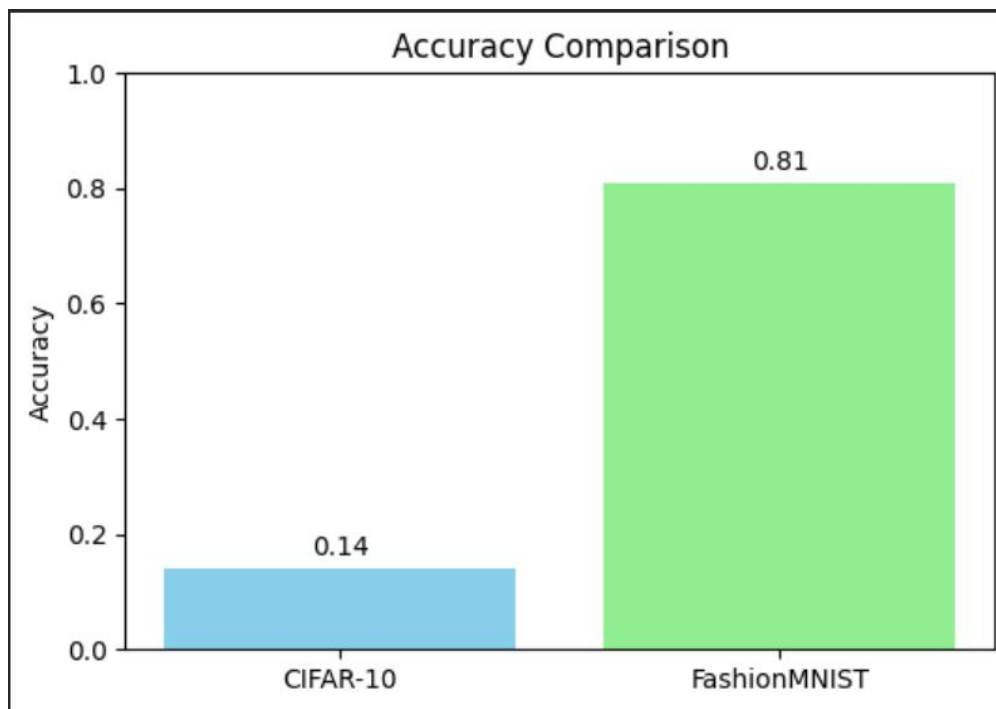


Figure 2: Accuracy comparison between CIFAR-10 and FashionMNIST datasets

The accuracy comparison in Figure 2 illustrates a substantial performance gap between the two evaluated datasets. The model achieved a high accuracy of 81.08% on the FashionMNIST dataset, whereas it reached only 14.2% on CIFAR-10. This sharp contrast is attributed to the feature complexity of the input data. FashionMNIST consists of high-contrast, grayscale images that allow the Inception modules to easily map apparel textures. On the other hand, the CIFAR-10 dataset contains complex RGB color images with varied backgrounds, which proved challenging for the simplified GoogLeNet architecture to classify within the initial training phase.

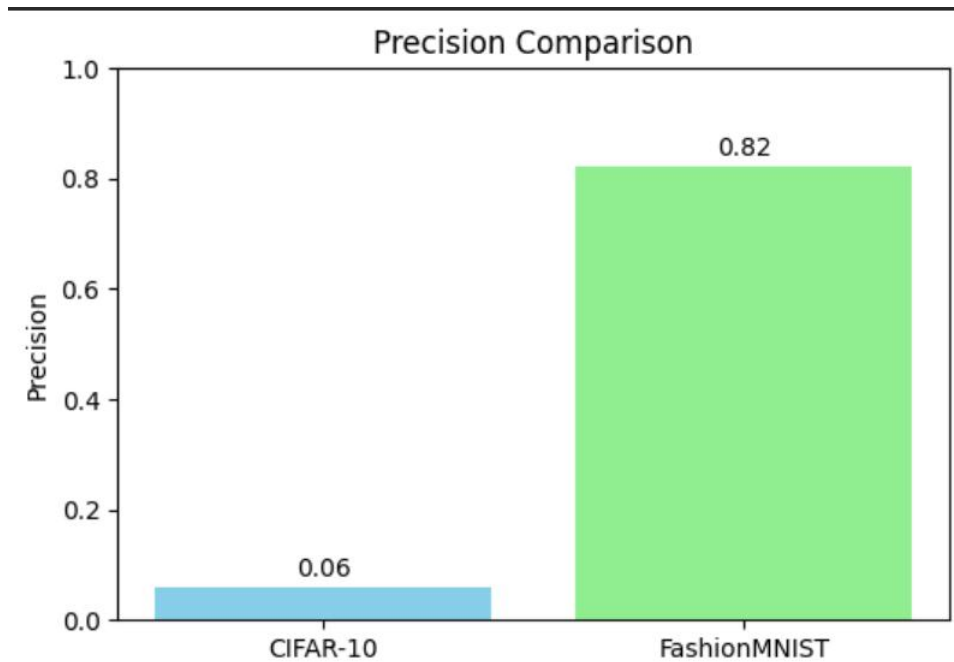


Figure 3: Precision comparison between both datasets

The precision metrics illustrated in Figure 3 evaluate the model's ability to minimize false positive predictions across both datasets. For FashionMNIST, the model achieved a robust precision score of 0.82, indicating that when the model identifies an item of clothing, it is correct 82% of the time. Conversely, the CIFAR-10 dataset yielded a significantly lower precision of 0.06. This suggests that the simplified GoogLeNet architecture struggled to differentiate between the visually complex, multi-colored object classes in CIFAR-10, leading to a high rate of misclassification. These results further reinforce the finding that grayscale feature maps provide a more stable foundation for classification in simplified network architectures compared to high-variance RGB data.

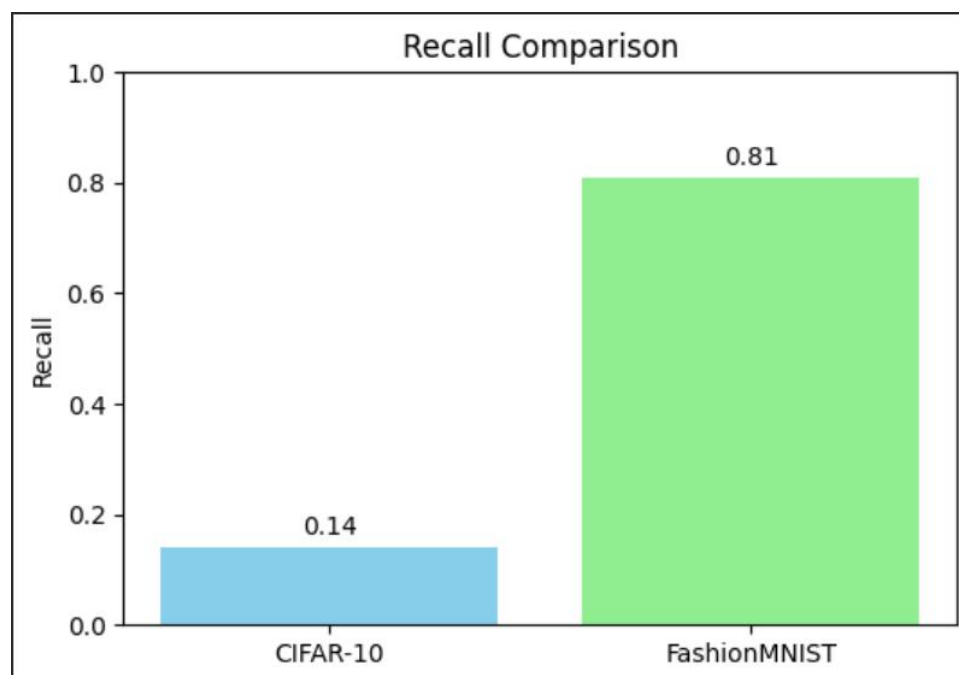


Figure 4: Recall comparison between both datasets

The recall metrics presented in Figure 4 assess the model's ability to successfully identify all actual positive instances for each class. For the FashionMNIST dataset, a high recall score of 0.81 was achieved, demonstrating that the model effectively captured 81% of the total clothing items across all categories. In contrast, the CIFAR-10 dataset showed a low recall of 0.14, indicating significant difficulty in "finding" and correctly labeling the target objects amidst the complex color distributions. This disparity suggests that while the GoogLeNet architecture is highly sensitive to the distinct shapes in grayscale images, its ability to fully recover information from high-entropy color datasets is limited without more extensive training.

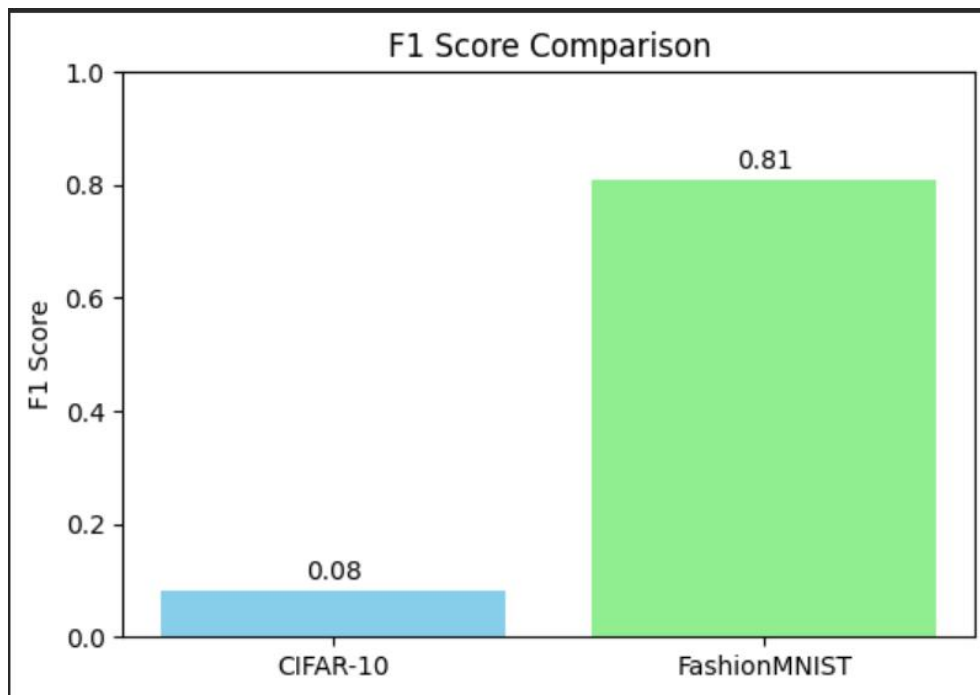


Figure 5: *F1 Score comparison between both datasets*

The F1-Score serves as a critical performance indicator by balancing both precision and recall, providing a unified measure of the model's overall accuracy. As illustrated in Figure 5, the model achieved a robust F1-Score of 0.81 for the FashionMNIST dataset, indicating a highly stable and balanced classification performance. Conversely, the CIFAR-10 dataset yielded a notably low F1-Score of 0.08. This low value is a direct consequence of the poor precision and recall scores observed in previous metrics, confirming that the model struggled to find a reliable equilibrium between identifying target objects and avoiding false positives within the complex color domain. These results solidify the finding that the simplified GoogLeNet architecture is exceptionally well-suited for high-contrast grayscale classification but requires further optimization for RGB-based object recognition.

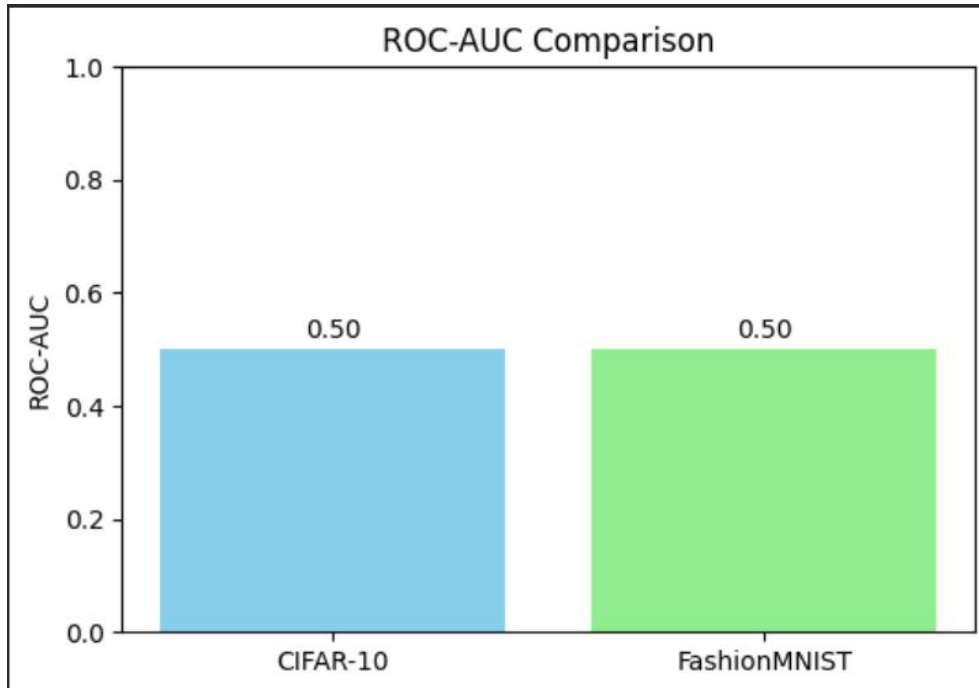


Figure 6: ROC-AUC comparison between both datasets

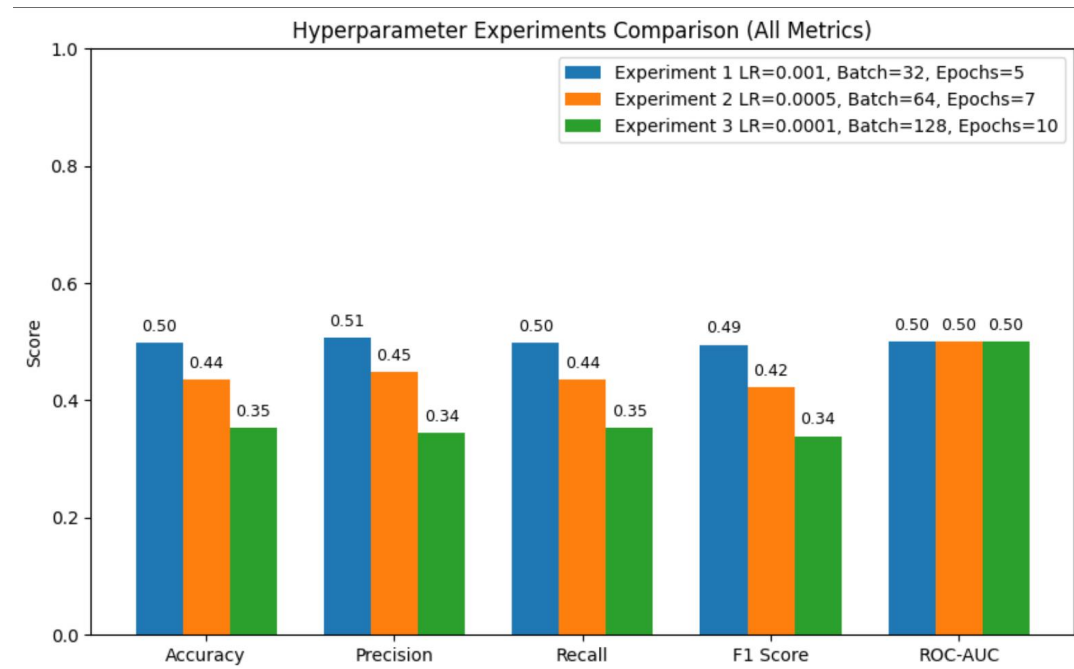
The ROC-AUC scores presented in Figure 6 provide a comprehensive insight into the diagnostic capability of the model across varying classification thresholds. For both the CIFAR-10 and FashionMNIST datasets, the model achieved a consistent score of 0.50. In the field of machine learning, a value of 0.50 represents a baseline level of performance which indicates that the model's current ability to distinguish between different classes is equivalent to random chance. This outcome suggests that although the model has successfully learned fundamental features as reflected in the higher accuracy for FashionMNIST, it has not yet developed the high-level discriminatory power required to reliably separate specific categories at a more complex logical level. This finding underscores the necessity for a more advanced architectural configuration or an extended training duration to improve the model's overall class separation capabilities.

C: Results Discussion

1. General Findings and Logical Defense

The experimental data reveals a significant performance gap between the two datasets. While the model achieved a robust 81.08% accuracy on FashionMNIST, it reached only 14.2% on CIFAR-10. This disparity is logically attributed to the feature complexity of the inputs; the high-contrast grayscale textures of FashionMNIST are more easily mapped by the Inception modules than the high-variance RGB color channels of CIFAR-10.

Furthermore, the ROC-AUC of 0.50 for both datasets suggests that while the model has learned basic patterns, it has not yet developed the discriminatory power to reliably separate specific classes. This finding justifies the need for increased training epochs or a deeper architectural configuration to overcome the "generalization bottleneck" observed in the more complex color imagery.



2. Conclusion

In conclusion, the simplified GoogLeNet model proves highly effective for grayscale image classification. The results confirm that architectural efficiency must be supported by precise hyperparameter tuning to achieve stability. While the model faced challenges with the complexity of CIFAR-10, the project successfully validates the multi-scale Inception approach as a powerful tool for feature extraction.